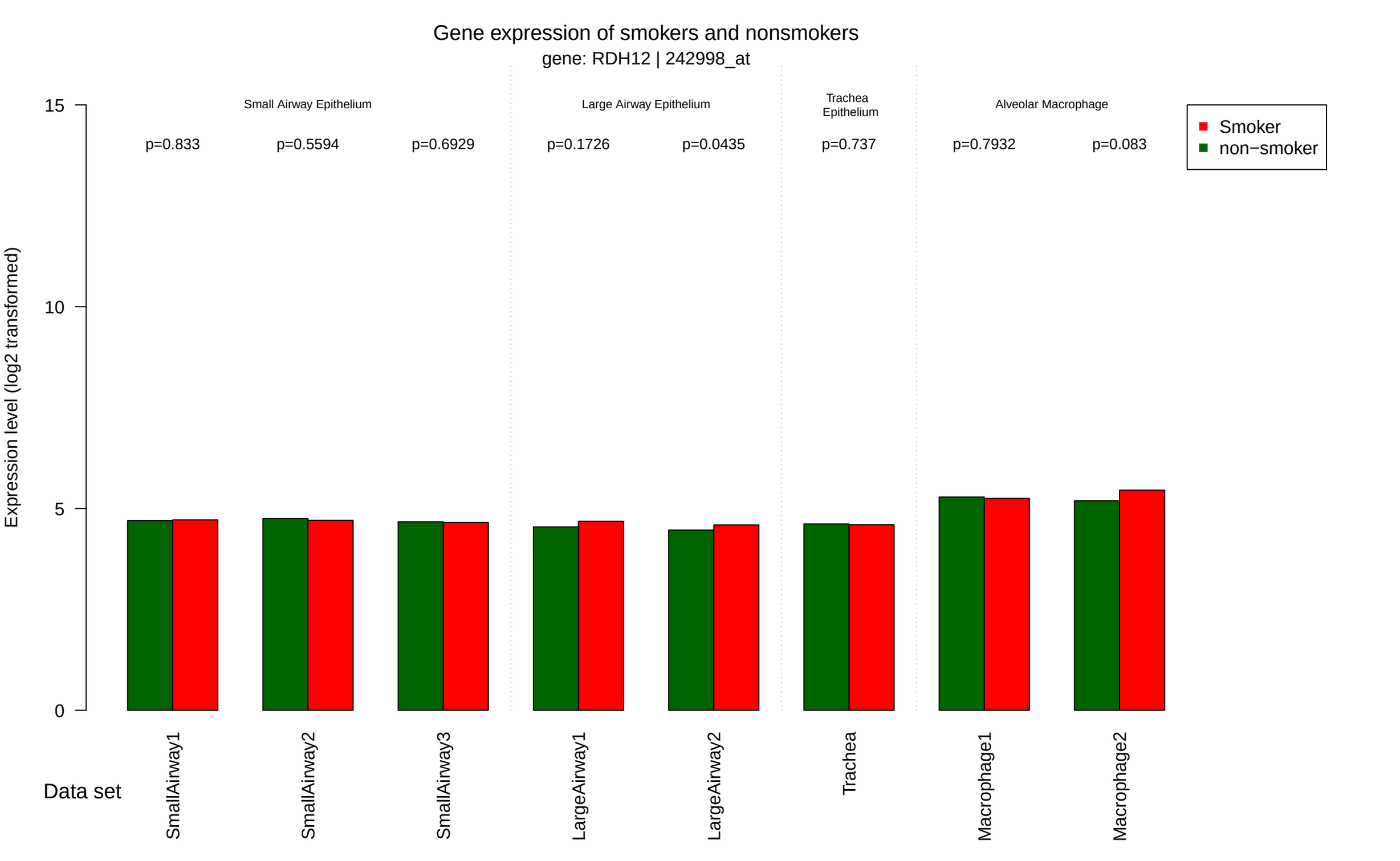
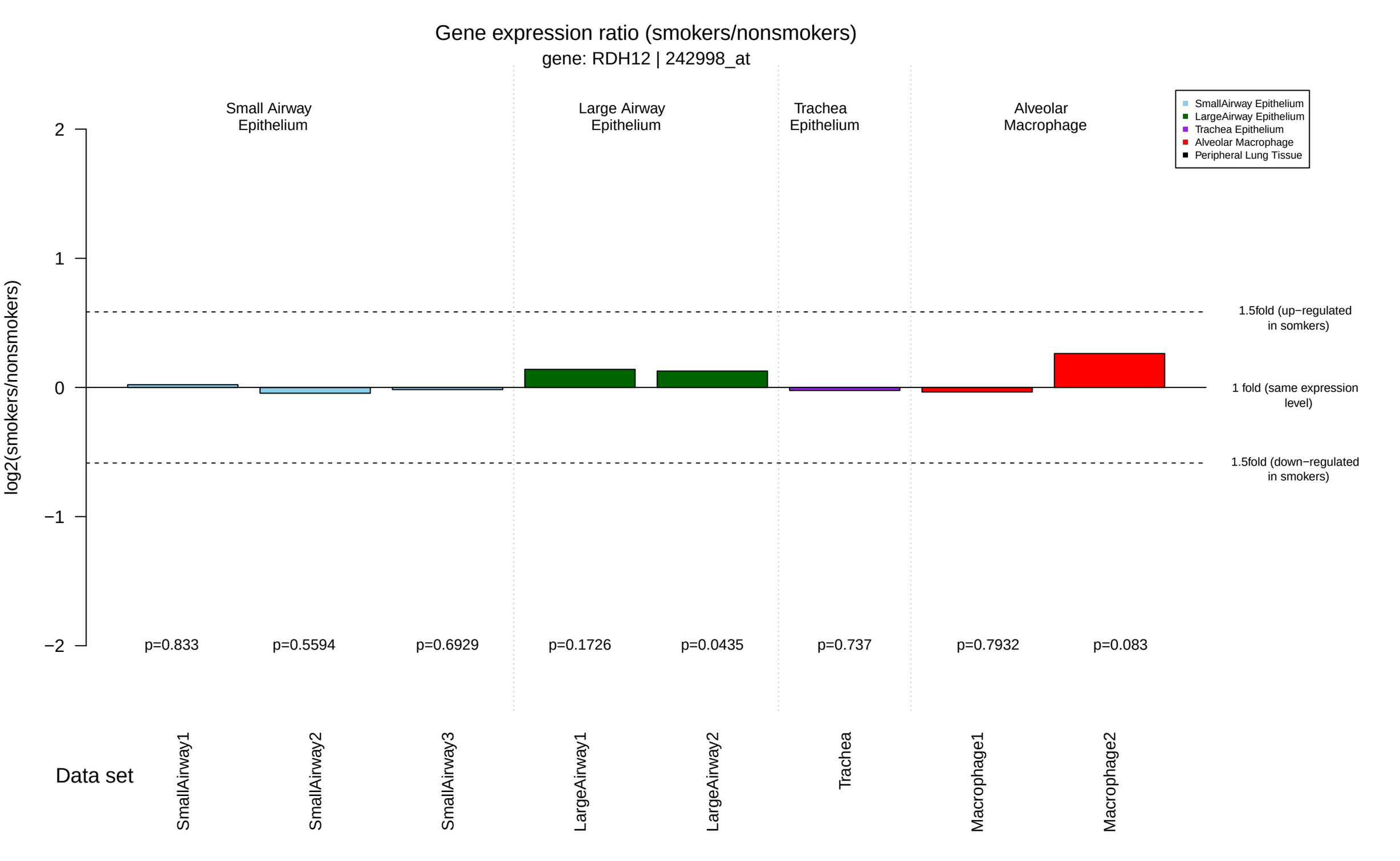
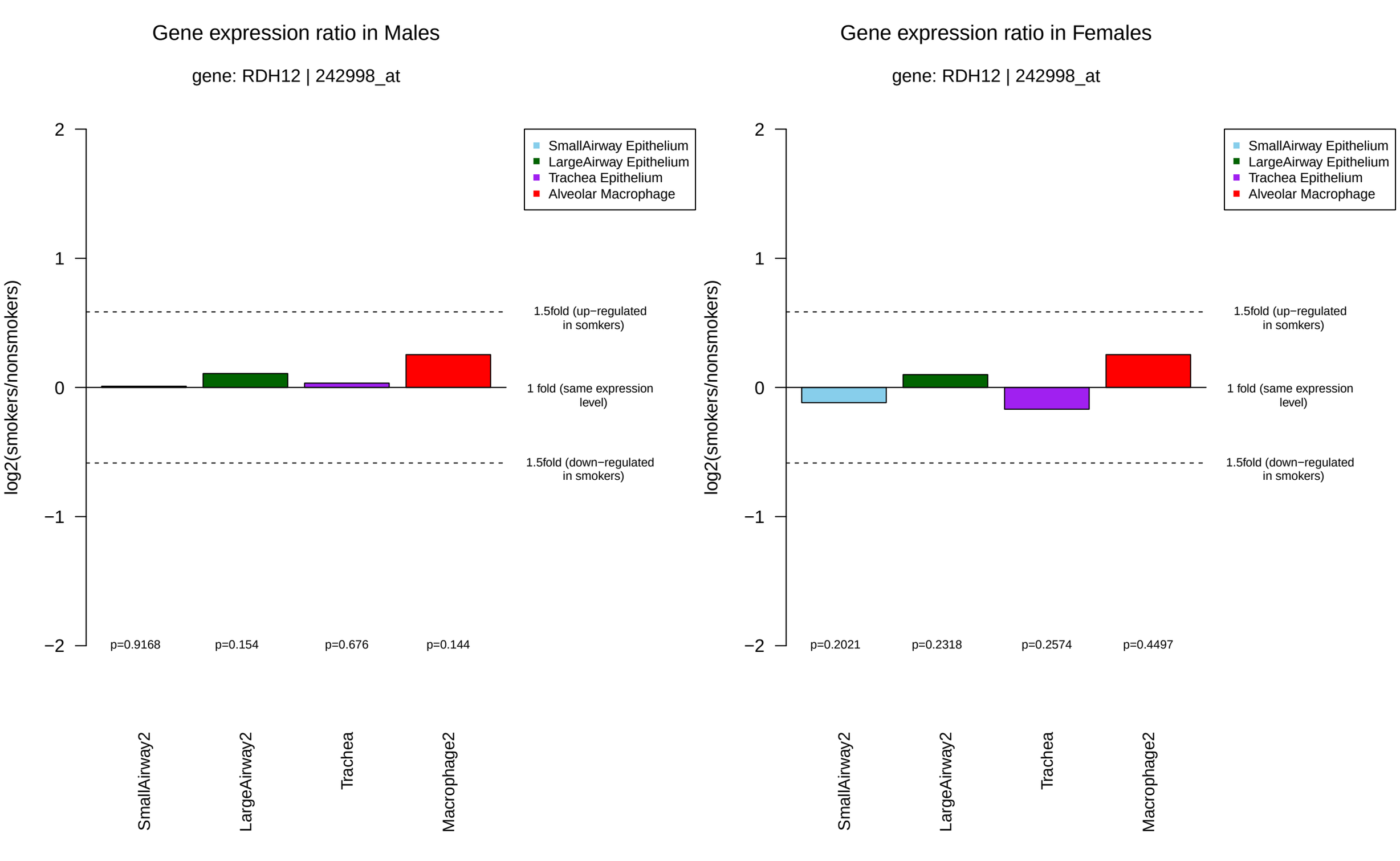
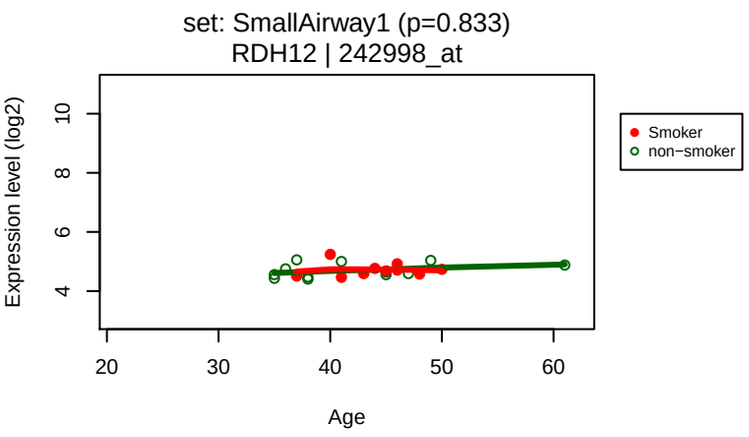
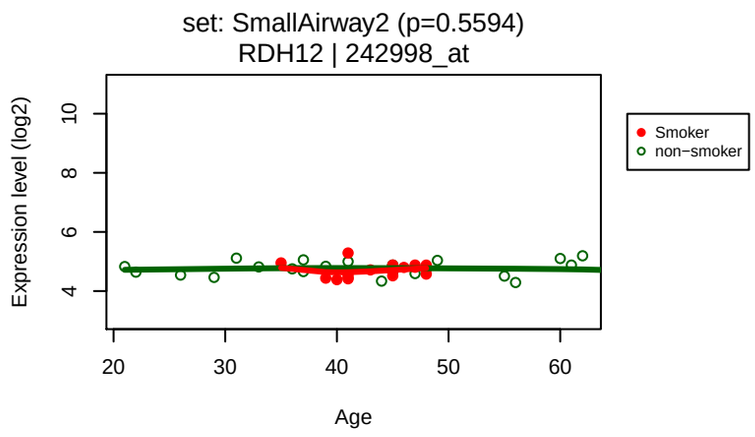
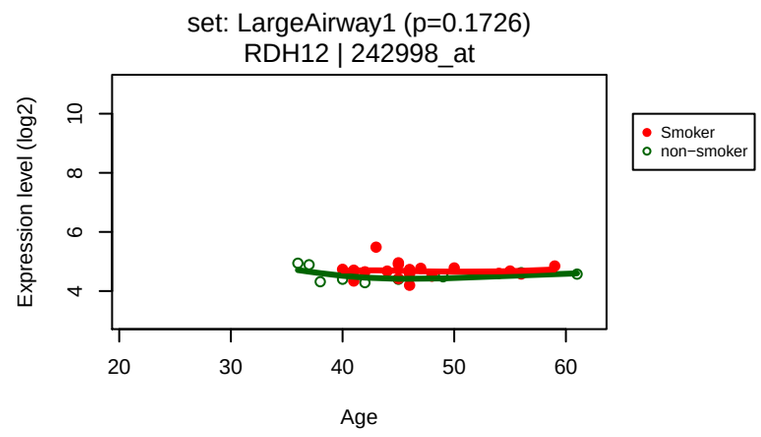

Gene expression of smokers and nonsmokers
gene: RDH12 | 242998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.833
p=0.5594
p=0.6929
p=0.1726
p=0.0435
p=0.737
p=0.7932
p=0.083
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

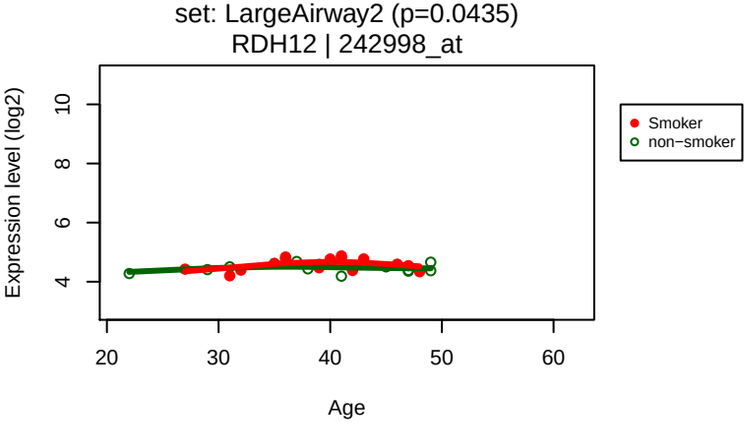

Gene expression ratio (smokers/nonsmokers)
gene: RDH12 | 242998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.833
p=0.5594
p=0.6929
p=0.1726
p=0.0435
p=0.737
p=0.7932
p=0.083
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

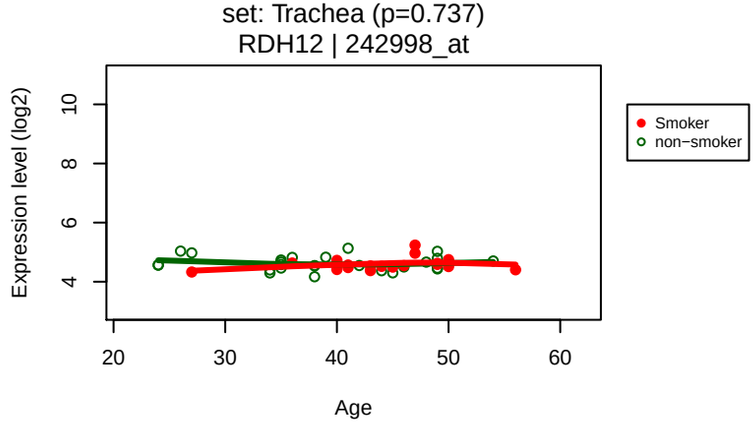

Gene expression ratio in Males
gene: RDH12 | 242998_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9168
p=0.154
p=0.676
p=0.144
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RDH12 | 242998_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2021
p=0.2318
p=0.2574
p=0.4497
SmallAirway2
LargeAirway2
Trachea
Macrophage2

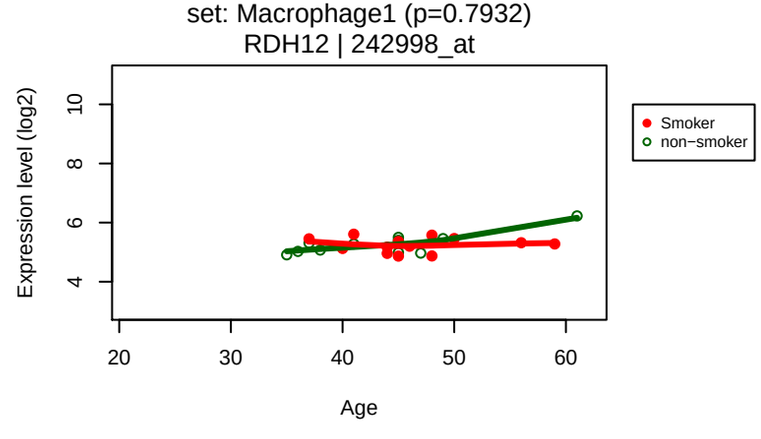

set: SmallAirway1 (p=0.833)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker

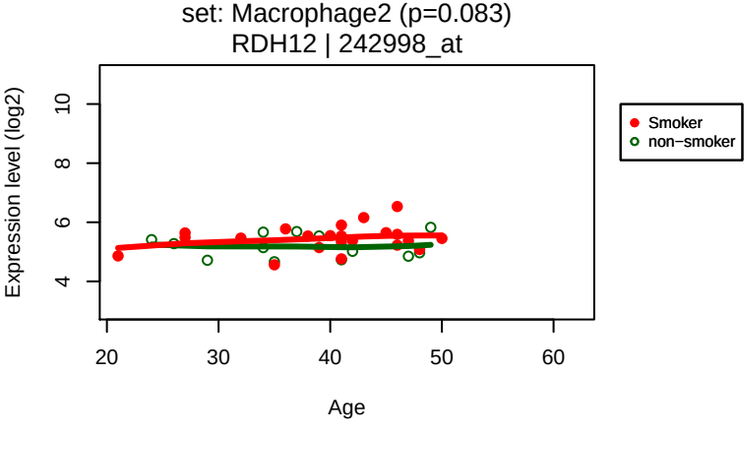

set: SmallAirway2 (p=0.5594)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker

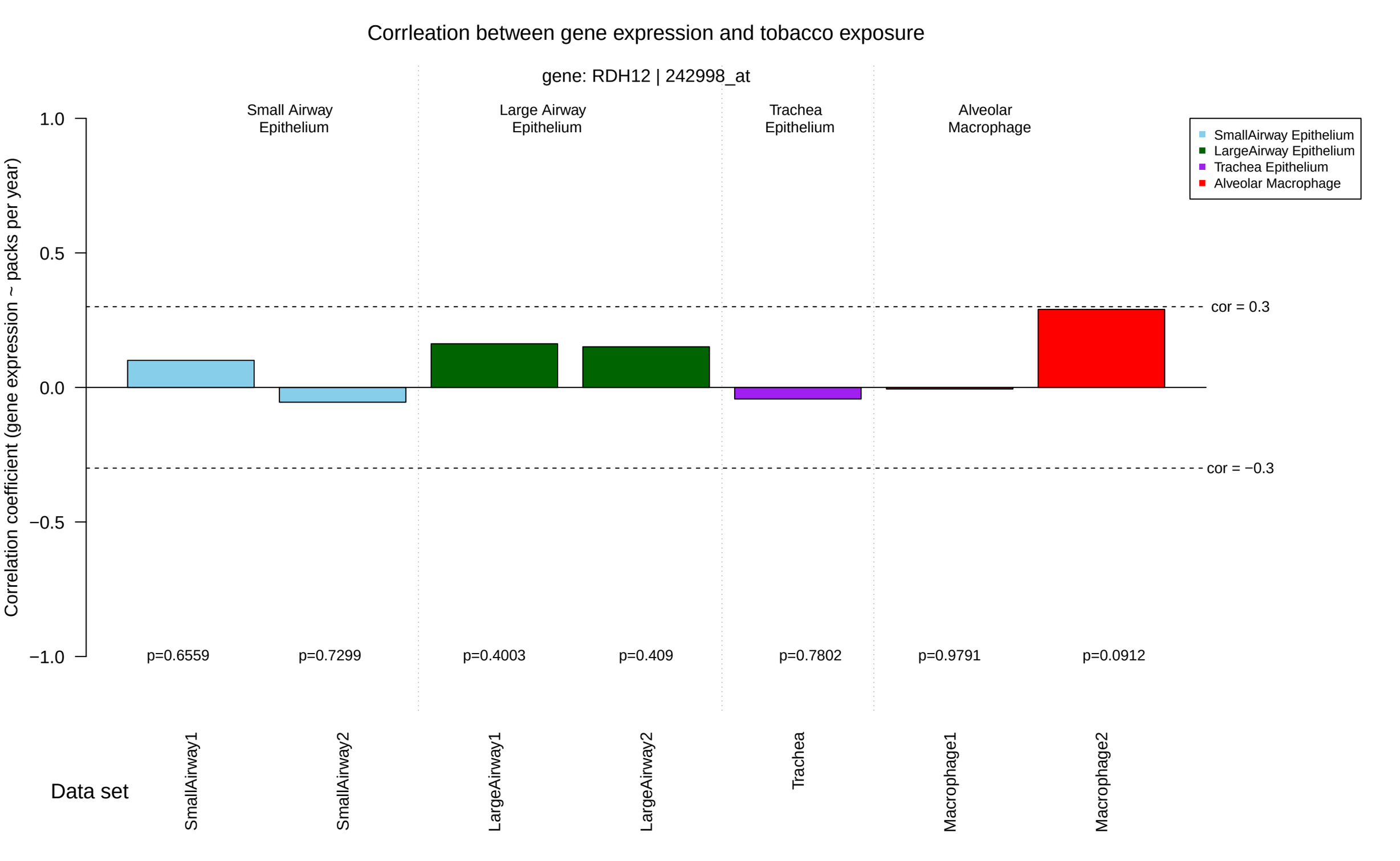

set: LargeAirway1 (p=0.1726)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0435)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.737)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7932)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.083)
RDH12 | 242998_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RDH12 | 242998_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6559
p=0.7299
p=0.4003
p=0.409
p=0.7802
p=0.9791
p=0.0912
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2